

OUTIL TRAMES

Line Halftone Tool

User Manual

Version 1.1

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Original algorithm: Jean-Noël Lafargue

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1. Overview

Outil Trames is a web-based artistic halftone tool using directed line hatching. It converts a source image into a network of hatching lines whose density and direction reproduce the tonal values of the original. The result can be exported as a multi-layer SVG compatible with Inkscape, or as a PNG for direct use.

Designed for screen printing, engraving, risography and any printing technique where line-based halftoning is preferable to traditional dot screens.

♦ How to use

Outil Trames is a standalone HTML file that opens directly in a web browser (Chrome, Firefox, Edge). No installation required. For the best experience, open the file in full screen (F11 on Windows/Linux, Cmd+Ctrl+F on Mac) — the interface is designed to occupy the full screen width and all four columns are only fully visible in full screen mode.

Language selection

The interface is available in French and English. The FR and EN buttons appear at the bottom of the left column, below the credits. The choice is automatically saved in the browser and will persist on future sessions.

- FR — French interface (default language)
- EN — English interface

Language switching is instant and applies to all labels, buttons, sections and options. Numeric values and current parameters are not affected.

Credits and License

Outil Trames was developed by Martial Rossignol. The line halftone algorithm is based on the work of Jean-Noël Lafargue.

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Interface — four columns

Left Column	Image Column
Open / image info	Interactive crop (popup)
Output format (series, number, orientation)	Tone controls (brightness, contrast, gamma)
Layout (margin, frame, target area)	Levels (black/white input)
H and V position	Gray range
SVG / PNG export	Gaussian blur / Sharpness
Color/Mono preview	
Language selector	
Central Column	Processing Column
Interactive preview	Halftone parameters (layers, thresholds, spacing, angle)

Scroll wheel zoom (cursor-centered)	Line width
Click-drag pan	Line wobble
Zoom indicator	JSON parameter save/load
Status bar	Generated layers list (visibility)

2. Recommended Workflow

Follow these steps in order for optimal results and to avoid unnecessary recalculations. Each step builds on the previous one.

♦ Before you start — full screen required

Open the `outil_trames.html` file in your browser, then switch to full screen (F11 on Windows/Linux, Cmd+Ctrl+F on Mac). The interface uses four side-by-side columns: without full screen, the right columns may be partially hidden and sliders inaccessible. The software requires no internet connection — everything runs locally in the browser.

Step 1 — Load the image

- Click  Open image or drag and drop directly into the central area.
- Accepted formats: PNG, JPG, GIF, WEBP.
- The original image appears in the Image column for cropping.

♦ Quality tip

Use a grayscale image when possible — results are more predictable. Color images are automatically converted to luminance, but pre-converting to grayscale in an external editor often gives better results.

Step 2 — Crop (optional)

- Click  Crop... to open the full-screen crop popup.
- Choose an aspect ratio: Free, Original, 1:1, 3:2, 2:3, 4:3, 3:4, 16:9.
- Draw the selection diagonally (top-left → bottom-right). The ratio is automatically enforced.
- Click  Apply — the preview updates immediately.
-  Full image cancels the crop and returns to the full image.

Step 3 — Image pre-processing

Pre-processing is applied to pixels before the halftone calculation. Work in this order:

- **Levels:** adjust Black and White input to stretch the histogram.
- **Gamma:** correct midtones. Values > 1 lighten midtones (more detail in highlights).
- **Gray range:** positive value stretches the range (more local contrast), negative compresses toward mid-gray.
- **Highlight comp.:** non-linearly compresses highlights — recovers detail in overexposed areas.
- **Brightness / Contrast:** global adjustments, apply last after levels.
- **Blur:** light Gaussian blur (1–2) softens transitions for smoother hatching.
- **Sharpen:** accentuates edges for crisper lines in detailed areas.

Step 4 — Output format

- Select the series (A, B or C), number (0 to 6) and orientation (Portrait/Landscape).

- Set the white margin (space between paper edge and drawing).
- Frame thickness creates a border around the drawing area (0 = no frame).
- Target area: if the drawing should occupy less than the full paper (e.g. A4 in an A3), choose the target format.
- H / V position: place the drawing within the available area.

Step 5 — Halftone parameters

- **Layers (1–12):** number of hatching directions. 1 = simple hatching, 2–4 = crosshatch, 6–12 = dense screen.
- **Min/Max threshold:** tonal range processed. Min = darkness level at which lines appear. Max = brightness level at which lines disappear.
- **Min/Max spacing:** distance between lines. Min = spacing in dark areas, Max = in light areas.
- **Start angle:** orientation of the first layer's hatching.
- **Line width:** in pixels on the preview. Default (0.8) gives fine lines suited to screen printing.

♦ Screen printing parameters

For clean screen printing: line width 0.8, min spacing 6, max spacing 20, 1 or 2 layers. Avoid angles close to 0° or 90° which create visible moiré effects.

Step 6 — Line wobble (optional)

- Line wobble: amplitude of the perpendicular oscillation. At 0: mechanically straight lines. Between 0.5 and 2: hand-drawn feel.
- Wobble freq.: number of oscillation cycles along a line's length. Value 2–3 for a natural effect.

♦ Subtle wobble

Very low values (amplitude 0.5–1.0, frequency 2) are enough to humanize the halftone without distorting it. Each segment's endpoints remain fixed — the halftone structure is preserved.

Step 7 — Layer management

- The generated layers list appears at the bottom of the Processing column.
- Click the eye icon to show/hide a layer in the preview. The layer remains in the SVG export.
- The number of lines per layer is shown — useful for evaluating density.

Step 8 — Export

- ↓ **Export SVG:** Inkscape-compatible multi-layer SVG. Each halftone layer is a separate group. Units are mm.
- ↓ **Export PNG:** captures the preview as displayed (screen resolution, with margins and frame).

3. Parameter Reference

Left Column — Format and Layout

Output format

- **Series A/B/C:** ISO series. A is the international standard. B and C are less common.
- **Number 0 to 6:** A0 (841×1189 mm), A1 (594×841 mm), A2 (420×594 mm), A3 (297×420 mm), A4 (210×297 mm), A5 (148×210 mm), A6 (105×148 mm). Same logic for B and C series.
- **Portrait / Landscape:** the preview canvas reflects the format change immediately.

Layout

- **White margin (0–50 mm):** uniform white space between paper edge and drawing area.
- **Frame width (0–10 mm):** black border around drawing area. 0 = no frame.
- **Target area:** surface in which the drawing is inscribed. Options: Full page, ½ width, ½ height, or a fixed ISO format.
- **H / V position:** placement of the drawing within the available space.

Language

- **FR / EN buttons:** switch the interface language instantly. The choice is saved in browser localStorage and persists across sessions.

Image Column — Pre-processing

Crop

The popup window displays the image at large size. The selected ratio constrains the selection. After applying, the preview recalculates automatically.

Tone

- **Brightness (–100/+100):** global additive shift. Use sparingly after levels.
- **Contrast (–100/+100):** multiplicative amplification around mid-gray.
- **Highlight comp. (0/100):** non-linear compression of highlights. Useful to recover detail in overexposed areas.
- **Gamma (0.1/3.0):** power curve correction. 1.0 = neutral. > 1 lightens midtones.

Levels

- **Black input (0/254):** pixels darker than this value are remapped to 0.
- **White input (1/255):** pixels lighter than this value are remapped to 255.
- **Gray range (–100/+100):** stretches (+) or compresses (–) the tone range around mid-gray.

Sharpness / Blur

- **Blur (0/10):** separable Gaussian blur. Useful values: 1–3.
- **Sharpen (0/5):** Laplacian sharpening filter. Use after blur to define edges.

Processing Column — Halftone

Main parameters

- **Layers (1–12):** number of hatching directions. Each layer is offset by $180^\circ/n$ from the previous.
- **Min threshold (0.001–0.999):** luminosity value below which lines appear.
- **Max threshold (0.001–0.999):** luminosity value above which lines disappear.
- **Min spacing (2–100):** line spacing in dark areas (pixels).
- **Max spacing (2–100):** line spacing in light areas.
- **Start angle (0–180°):** direction of the first layer.
- **Line width (0.1–8):** line thickness in pixels on the preview.

Wobble

- **Line wobble (0–5):** perpendicular oscillation amplitude. 0 = straight lines. Endpoints remain fixed.
- **Wobble freq. (0.5–8):** number of oscillation cycles. Absolute spatial frequency — independent of segment length.

4. Tips and Best Practices

Source image preparation

- Convert to grayscale before loading if possible — results are more predictable.
- Avoid very low-resolution images: below 800×800 px, hatching lines lack definition.
- Images with high local contrast (portraits, engravings, illustrations) give the best results.
- Very flat photographs (low contrast) require aggressive levels pre-processing.

♦ Crop first

Crop the image BEFORE adjusting halftone parameters. The crop changes the image ratio, which influences how the drawing fits in the paper format.

Halftone parameter tuning

- Start with 2 layers, 45° angle, equal spacings (min = max = 15). Adjust from there.
- If light areas are too dense: increase max threshold or max spacing.
- If dark areas lack density: decrease min threshold or min spacing.
- A very low min spacing (3–5) in shadows gives a near-solid fill effect.

Line wobble

- Wobble humanizes the halftone but should remain subtle to preserve readability.
- Amplitude 0.5 + frequency 2–3: pen-line effect, almost imperceptible in print.
- Amplitude 2–3 + frequency 1: visible soft curves, woodcut engraving effect.
- Do not exceed amplitude 3 for screen printing — beyond that, overlapping areas create uncontrolled thickening.

Presets and reproducibility

- ↓ Params saves the 7 halftone parameters to a JSON file.
- ↑ Params reloads a preset and automatically re-renders.
- Name JSON files with key parameters: e.g. portrait_2layers_45deg_sp15.json.
- Pre-processing parameters are not saved in JSON — note values separately if needed.

5. Shortcuts and Interactions

Central preview

Action	Effect
Scroll wheel	Zoom centered on cursor
Click-drag	Pan the preview
Full screen button	Toggle full screen mode
FR / EN buttons	Switch interface language

Crop popup

Action	Effect
Diagonal click-drag	Draws the selection area
Choose a ratio before drawing	Constrains selection to ratio
Escape key	Close without applying
Click on background	Close without applying

6. SVG Export Structure

The exported SVG file is designed for direct integration into Inkscape without further manipulation.

Layers and namespaces

- inkscape: and sodipodi: namespaces included for maximum compatibility.
- document-units='mm' in namedview: Inkscape opens the document in millimeters.
- Background layer: white rectangle the size of the paper.
- Halftone layers 1–N: one group per direction, with the angle in degrees in the name.
- Frame layer: border rectangle, modifiable independently.
- Layers hidden in preview: exported with display='none', recoverable in Inkscape.

♦ Modifying in Inkscape

To change a halftone layer color: open the Layers panel, select the layer, then Object > Object Properties > Stroke color. The color applies to all elements in the layer simultaneously.